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Designing and validating a mathematics learning model integrating green economy and numeracy for ESD

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Abstract

Integrating sustainability into primary mathematics education remains a significant challenge in Indonesia, particularly in aligning numeracy learning with Education for Sustainable Development (ESD) and green economy principles. This study aimed to design and validate the Green Economy and Numeracy Integration for Understanding Sustainability (GENIUS) learning model, a structured framework intended to integrate sustainability and green economy principles into primary mathematics education through contextualized, reflective, and student-centered instructional practices. Guided by constructivist learning theory, contextual learning, and the ESD competency framework, a mixed-methods design recruited 78 unique primary school teachers in Bogor City, West Java, selected through purposive sampling. Data were collected through questionnaires ($n=52$), semi-structured interviews ($n=15$), and classroom observations ($n=8$), with overlapping participation across instruments, to examine teachers' needs and instructional constraints. The needs analysis revealed that although most teachers showed positive attitudes toward sustainability-oriented instruction, major barriers included limited ESD understanding, rigid curriculum structures, and insufficient contextual resources. Based on these findings, the GENIUS model was developed with components including learning objectives, instructional syntax, roles of teachers and students, support systems, and contextual materials. Expert validation by mathematics education experts, ESD specialists, and experienced primary teachers used Aiken's V to assess content and construct validity. Results indicate high validity across components ($V \geq 0.85$), exceeding accepted thresholds and indicating coherence and practical feasibility. This study demonstrates the novelty of the GENIUS model for ESD numeracy learning.

Keywords Mathematics learning model, Green economy, Numeracy, Sustainable education, Model validation

1 Introduction

Global challenges such as climate change, environmental degradation, and economic inequality have intensified the demand for education systems that prepare learners to engage with sustainability-oriented problems [37, 45]. Education for Sustainable

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Development (ESD) provides a transformative approach to cultivating learners' competencies for sustainable living [24]. Rather than functioning as an additional subject, ESD calls for the systematic integration of sustainability principles across curricular areas to enable learners to connect knowledge with real-world challenges.

Prior research emphasizes the role of ESD integration in curricula as a key driver for nurturing critical thinking, systems literacy, and global citizenship [20]. Embedding sustainability principles across subjects empowers students to connect theoretical learning with real-world issues [10]. Mathematics, although often considered a neutral discipline, holds strategic potential in this context due to its strong focus on analytical reasoning, quantitative literacy, and problem-solving skills [58]. However, this potential remains underutilized when mathematics is taught primarily as an abstract and decontextualized discipline.

Recent studies have explored interdisciplinary models linking mathematics learning with environmental education and sustainability goals [7, 27, 30]. Additionally, technological innovations such as adaptive learning platforms and XR-based approaches have emerged as promising tools to reinforce ESD integration [46, 50]. These developments indicate a growing recognition of the role of mathematics education in fostering sustainable mindsets and practices.

In Indonesia, the government has introduced the *Merdeka Curriculum* to promote more flexible and context-based learning [53]. However, the explicit integration of sustainability principles, particularly within mathematics education, remains limited [29]. Environmental and sustainability issues are predominantly addressed in science or social studies, while mathematics instruction continues to focus on procedural fluency with limited real-world relevance [25]. This disconnect undermines the curriculum's broader goal of character development and contextual learning, leaving students insufficiently prepared to apply mathematical reasoning to sustainability-related challenges.

Evidence from large-scale assessments highlights these challenges. Results from the National Assessment and international studies such as PISA consistently show that Indonesian students struggle with higher-order numeracy and contextual problem-solving [56]. While they may perform adequately in procedural calculations, applying mathematics to sustainability-related scenarios such as energy consumption, waste reduction, or resource management remains underdeveloped [47]. This reflects the absence of structured models that embed sustainability into numeracy leasrning at the primary level.

Despite its potential, mathematics education at the primary level both globally and in Indonesia remains heavily focused on procedural tasks and abstract exercises that fail to connect with sustainability contexts [33, 54]. Contextual approaches such as numeracy assessments and gamification have shown incremental improvements [28, 39, 44], but their impact on sustainability awareness and critical thinking is limited. Moreover, the literature suggests that ESD-oriented pedagogy is primarily implemented in teacher education or higher levels of study, with little evidence at the primary level [8, 21, 51]. Explicit models integrating numeracy with green economy principles are also scarce including in Indonesia.

To address this gap, the present study proposes the GENIUS model, an original mathematics learning framework specifically developed for this research. The model was conceptually grounded in ESD principles and designed to integrate numeracy and green

economy concepts through contextualized, reflective, and student-centered learning activities. Its development process involved a needs analysis of primary mathematics teachers, iterative design based on pedagogical theory, and expert validation to ensure content, construct, and practical validity. The model is theoretically grounded in constructivist and contextual learning theories, as well as UNESCO's ESD competency framework, which emphasizes systems thinking, values-based reasoning, and action-oriented learning. GENIUS operationalizes these principles through a structured instructional syntax that situates mathematical concepts within sustainability contexts, such as energy consumption, sustainable consumption patterns, and waste management—key dimensions of the green economy [32, 55].

The novelty of this research lies in embedding sustainability-oriented values and competencies directly into the core syntax of mathematics instruction rather than treating sustainability as an add-on theme. Unlike conventional mathematics teaching, which prioritizes procedural mastery, or existing ESD approaches that rely on thematic integration, the GENIUS model aligns learning objectives, instructional activities, and assessment with sustainability-driven numeracy tasks. This approach positions mathematics as a tool for responsible decision-making and sustainable living, addressing both cognitive and value-based learning outcomes. Within the Indonesian context, the model offers a concrete pedagogical response to curriculum reform demands and persistent numeracy challenges identified in national and international assessments.

Accordingly, the primary objective of this study is to design, develop, and validate the GENIUS model as a pedagogical framework for primary mathematics education. Specifically, the study aims to: (1) identify contextual needs and challenges in integrating ESD into primary mathematics learning, (2) develop a mathematics learning model that embeds numeracy within green economy and sustainability contexts, and (3) examine the model's validity in terms of content, construct, and practicality through expert evaluation. Based on these objectives, the following research questions are addressed:

RQ1: What are the contextual needs and challenges in integrating sustainability principles into primary mathematics education, particularly in Indonesia?

RQ2: How can a mathematics learning model be designed to embed green economy concepts within the ESD framework?

RQ3: To what extent is the proposed GENIUS model valid in terms of content, construct, and practicality for classroom implementation?

2 Literature review

2.1 Learning models

Learning models are conceptual frameworks that organize learning experiences and guide the design, implementation, and evaluation of teaching [23]. They provide more than teaching methods; they offer a systematic structure with theoretical rationale, clear objectives, structured syntax, social systems, support mechanisms, and success criteria [19, 49]. Salama and Patil, [48] highlight the role of learning models in fostering social dynamics and collaboration, while Oviedo Ramirez et al., [38] emphasize alignment with digital and adaptive learning systems. Bilal and Eriști, [5] and An et al., [3] underscore interdisciplinary and self-directed aspects, particularly within project-based or STEAM contexts. Nel and Bornman, [36] further stress inclusivity, ensuring models

accommodate diverse learner needs. Kostaki & Linardakis [24] highlight inquiry-driven frameworks that connect disciplines with real-world activities.

Despite their strengths, existing learning models rarely integrate sustainability or ecological awareness explicitly. Most focus on cognitive and procedural outcomes, leaving values, environmental consciousness, and social responsibility underdeveloped. This gap indicates the need for a learning model, like GENIUS, that embeds sustainability competences directly into mathematics instruction.

2.2 Green economy

The green economy integrates economic, social, and environmental dimensions for sustainable development [2]. Febriansyah et al., [13] emphasize its focus on resource efficiency and reducing environmental impacts. Zayyadi, [57] frames green economy through principles of justice and ethical responsibility, while Ghasemi et al., [15] describe it as grounded in circular economy principles and green finance, such as green bonds. Nana et al., [35] highlight industrial ecosystem transformation toward circular business 4.0.

While green economy is conceptually rich, its application in primary mathematics instruction remains minimal. There is a gap in translating sustainability concepts into classroom mathematics, which could contextualize numeracy through energy management, sustainable consumption, or resource efficiency. This gap supports the rationale for integrating green economy principles into the GENIUS model.

2.3 Numeracy literacy in the context of minimum competence assessment

Numeracy literacy connects mathematical knowledge with practical, real-life problem solving. *Kemdikbudristek* defines it as the ability to use mathematical concepts, procedures, facts, and tools to solve daily problems [42]. Similarly, OECD/PISA emphasizes formulating, applying, and interpreting mathematics across contexts [4]. Bulut and Borromeo Ferri, [7] argue numeracy links mathematics with real-world situations, while Corres et al., [9] highlight its flexible and reflective application. Díez-Palomar et al., [11] stress its use in informed decision-making based on data.

In practice, primary mathematics instruction often focuses on procedural fluency, with limited connection to sustainability issues. Integrating numeracy with green economy and ESD contexts can enhance problem-solving, critical thinking, and environmental awareness, providing the rationale for GENIUS's contextualized approach.

2.4 Education for sustainable development (ESD)

ESD aims to transform learners' knowledge, attitudes, and skills for sustainable development. UNESCO [18]. defines ESD as empowering learners to make responsible, future-oriented decisions ensuring environmental, economic, and social sustainability. Fernando and Tajan, [14] emphasize connecting environmental, social, and economic knowledge in curriculum design. Price et al., [41] highlight ESD as fostering critical, collaborative, and reflective competencies. Glavič, [17] frames ESD as a global instrument embedding sustainability into education, while Giangrande et al., [16] link it to SDG 2030 objectives.

Despite its recognized benefits, ESD integration in mathematics particularly at the primary level is limited. Current practice often restricts sustainability education to

science or social studies, leaving mathematics as abstract and context-neutral. This gap underscores the need for models like GENIUS that combine numeracy, sustainability, and values education in a coherent framework.

2.5 Conceptual synthesis

Synthesizing these four constructs clarifies the theoretical rationale for GENIUS. Learning models provide structural and pedagogical guidance; green economy offers authentic sustainability contexts; numeracy literacy ensures mathematical reasoning applies to real-world problems; and ESD embeds critical thinking, reflective skills, and responsible decision-making. Together, they justify a learning model that simultaneously develops numeracy skills and sustainability awareness. The absence of existing models integrating these elements highlights a significant research gap that GENIUS seeks to address, ensuring primary mathematics education is both cognitively and socially meaningful.

3 Research methodology

This study employed a Research and Development (R&D) approach following the Borg and Gall model [6, 31, 40]. Given the scope and time constraints, the research was limited to three major stages: (1) needs analysis, (2) draft model design, and (3) expert validation; it is acknowledged that excluding full implementation and pilot testing may limit direct evidence of classroom effectiveness, which will be addressed in the discussion. The purpose of this study was to design and validate the GENIUS learning model—a mathematics learning framework that integrates numeracy and green economy concepts within the ESD paradigm. The adaptation of Borg and Gall's model was deemed appropriate as it systematically guides researchers in developing, testing, and refining educational innovations, making it particularly suitable for pedagogical model development.

3.1 Research context and participants

The study was conducted in selected primary schools in Bogor City, West Java Province, Indonesia, chosen due to their diverse student population and the local government's emphasis on sustainability education. Participants included primary school teachers with a minimum of five years of teaching experience and actively involved in teaching mathematics. Purposive sampling was applied based on specific inclusion criteria: teachers must be currently teaching mathematics, have at least five years of experience, and be willing to participate in multiple data collection stages. Schools were selected to represent a variety of resource levels and student demographics to enhance transferability. A total of 78 unique primary school teachers were recruited for the study. Data collection involved multiple instruments. Of these teachers, 52 completed the questionnaire, 15 participated in semi-structured interviews, and 8 classrooms were observed. Several teachers took part in more than one data collection activity; therefore, these figures represent data collection instances rather than distinct individuals. In total, 75 teacher engagements were recorded across all instruments.

3.2 Instruments and data collection

Three instruments were employed:

(1) Questionnaire: Developed to capture teachers' perceptions, challenges, and needs regarding the integration of numeracy, ESD, and green economy into mathematics

learning. The instrument included both closed-ended items on a five-point Likert scale and open-ended questions to elicit richer responses. Measures to reduce self-report bias included anonymous responses and encouraging honest feedback.

(2) Semi-structured interviews: Conducted with a subset of participants to explore in-depth their teaching practices, perceived barriers, and expectations. Interview protocols were designed to capture qualitative nuances that might not emerge from the questionnaires. Interviews were coded by two independent researchers, and intercoder reliability was calculated (Cohen's kappa = 0.82).

(3) Classroom observations: Undertaken to document real teaching practices, focusing on teaching methods, media use, student engagement, and attempts to integrate sustainability concepts. Observations employed a structured checklist supplemented by descriptive field notes.

The instruments were adapted from previous validated studies [12, 22, 34, 43] and reviewed by two experts in mathematics education and educational measurement to ensure content validity. Reliability of the questionnaire was tested through a pilot study, yielding a Cronbach's alpha coefficient of 0.87, indicating acceptable internal consistency.

3.3 Needs analysis

The needs analysis aimed to identify:

- Teachers challenges in strengthening numeracy within primary mathematics education.
- Opportunities and barriers to embedding ESD and green economy concepts.
- Resources, training, and support required by teachers to implement such integration.

Triangulation of questionnaire, interview, and observation data was employed to strengthen validity. Coding and thematic analysis followed systematic procedures to ensure accuracy and minimize researcher bias.

3.4 Model design

Based on the needs analysis and literature review, an initial draft of the GENIUS model was constructed. The draft included: (a) the model's philosophical and theoretical rationale, (b) instructional goals, (c) learning syntax, (d) teacher and student roles, (e) social and support systems, and (f) supplementary learning tools (lesson plans, teaching modules, worksheets, media, and assessment instruments). The draft was visually presented in flowcharts and detailed documentation to facilitate expert evaluation.

3.5 Expert validation

The draft model was validated by a panel of experts consisting of three mathematics education specialists, two ESD experts, and three senior primary school teachers. Experts were selected based on academic qualifications, teaching experience, and prior engagement with sustainability education. Validation was conducted independently and followed by a consensus discussion to resolve any disagreements.

Validation included both Likert-scale rating sheets (1 = very weak, 5 = very strong) and qualitative feedback to improve content validity, theoretical soundness, and practicality.

3.6 Data analysis

A mixed-methods approach was employed to analyze data:

- Quantitative analysis: Expert validation ratings were analyzed using Aiken's V coefficient to determine the validity of each component. A threshold of V ≥ 0.80 was applied as the criterion for high validity [1]. Items below the threshold were revised or removed based on expert feedback.
- Qualitative analysis: Written expert comments, interview transcripts, and observation notes were analyzed thematically using iterative coding (open, axial, and selective coding) by two coders, with intercoder reliability assessment and software-assisted organization using NVivo 12.

3.7 Ethical considerations

The study obtained ethical approval from Universitas Pakuan. Informed consent was secured from all participants, ensuring their voluntary involvement. Confidentiality was maintained by anonymizing participant data, and participants were given the right to withdraw from the study at any stage.

Finally, Fig. 1 illustrates the research and development process, and all steps are thoroughly explained in the text, ensuring methodological transparency.

4 Results and finding

4.1 Overview of research context and participants

The study was conducted in the context of primary education in Bogor City, West Java Province, Indonesia, an area characterized by rapid urbanization, demographic diversity, and increasing attention to sustainability-focused curriculum initiatives. Bogor City was deliberately selected due to its distinctive educational profile: the local government has

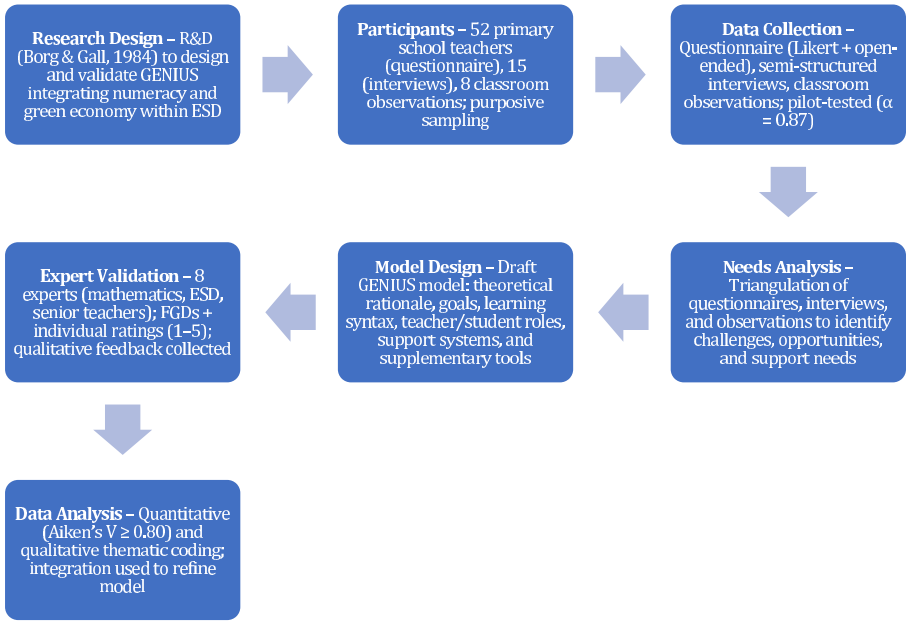


Fig. 1 Methodological flowchart – GENIUS learning model study

integrated environmental literacy and numeracy into the curriculum framework while also facing challenges related to teacher preparedness and pedagogical adaptation.

A total of 78 primary school teachers were purposively recruited from seven public and private schools. Selection criteria included a minimum of five years of teaching experience, active involvement in mathematics instruction, and familiarity with curriculum innovation projects initiated by the local education authority. Data were collected using multiple instruments. Of the 78 unique teachers, 52 completed the structured questionnaire, 15 participated in semi-structured interviews, and eight classroom observations were conducted across schools with varying socio-economic backgrounds. Several teachers participated in more than one data collection activity; therefore, these figures represent data collection instances rather than distinct individuals.

The participant profile indicated diversity in terms of age, years of service, and professional development exposure. Approximately 60% of respondents were female, and the majority (70%) were between 30 and 45 years old, with an average teaching experience of 11.6 years. Only 28% of teachers had previously attended formal training programs specifically related to numeracy and sustainability education.

Classroom observations highlighted differences in school environments. Some schools were well-resourced with multimedia tools and flexible learning spaces, while others faced infrastructural limitations, including overcrowded classrooms and limited access to teaching materials.

In summary, the participant and contextual data provide a descriptive overview of the educational setting in Bogor City, which forms the basis for the subsequent subsections presenting questionnaire, interview, and classroom observation findings.

4.2 Questionnaire findings

The questionnaire was completed by 52 teachers and measured four dimensions: pedagogical knowledge for sustainability, numeracy integration skills, instructional strategies, and professional attitudes toward innovation, using a five-point Likert scale. The instrument demonstrated high internal consistency (Cronbach's alpha = 0.876), with item-level values ranging from 0.792 to 0.884.

Descriptive statistics for the four dimensions are presented in Table 1. Teachers scored highest on professional attitudes toward innovation (M = 4.12, SD = 0.61) and lowest on numeracy integration skills (M = 3.47, SD = 0.79). Instructional strategies and pedagogical knowledge for sustainability received intermediate scores (M = 3.89 and M = 3.68, respectively).

A one-way ANOVA was conducted to examine differences in teachers' scores across the seven schools. Results indicated a statistically significant difference in overall competence among schools ($F(6, 45) = 4.37$, $p < .01$). Post-hoc analyses using the Tukey HSD test were performed to determine which schools differed; these results are presented in

Table 1 Descriptive statistics of teacher questionnaire responses ($n = 52$)

Dimension	Mean	SD	Min	Max	Cronbach's α
Pedagogical knowledge for sustainability	3.68	0.74	2.1	4.9	0.884
Numeracy integration skills	3.47	0.79	1.9	4.8	0.861
Instructional strategies	3.89	0.66	2.3	5	0.834
Professional Attitudes toward innovation	4.12	0.61	2.7	5	0.792
Overall scale	3.79	0.7	2.1	5	0.876

Table 2. The data show variation in teacher responses across the four dimensions, with professional attitudes generally higher than practical numeracy integration skills.

4.3 Interview findings

Semi-structured interviews were conducted with 15 teachers to gain deeper insights into their pedagogical practices, challenges, and expectations regarding the integration of numeracy with sustainability and green economy principles. Thematic analysis was employed following an iterative process of open coding, axial coding, and selective coding [52]. Three overarching themes emerged: (1) Perceptions of Numeracy within ESD, (2) Barriers to Pedagogical Implementation, and (3) Expectations for Professional Development and Support.

Theme 1: Perceptions of Numeracy within ESD

Thirteen out of 15 teachers discussed numeracy primarily as basic arithmetic skills and problem-solving within real-life contexts. However, only five teachers demonstrated awareness of the potential to link numeracy with sustainability or green economy principles.

- "I can teach students how to calculate water usage, but I am not sure how that becomes part of the green economy or sustainability. The connection is still unclear for me." (Teacher 3).
- "When calculating money spent on energy, I mention conservation, but I do not have a framework to make it part of a sustainability lesson." (Teacher 8).
- "I often give word problems from the textbook. Occasionally I try to include environmental contexts, but I am not systematic about it." (Teacher 12).

Theme 2: Barriers to Pedagogical Implementation

Table 2 Post-hoc Tukey HSD comparison of teacher scores across schools

Comparison (School Pairs)	Mean difference	Std. Error	*p*-value	Significant
School A – School B	0.3	0.15	0.12	No
School A – School C	0.05	0.15	0.99	No
School A – School D	0.5	0.15	0.04	Yes
School A – School E	−0.6	0.15	0.02	Yes
School A – School F	0.15	0.15	0.88	No
School A – School G	−0.5	0.15	0.05	Yes
School B – School C	−0.25	0.15	0.45	No
School B – School D	0.2	0.15	0.68	No
School B – School E	−0.9	0.15	0.001	Yes
School B – School F	−0.15	0.15	0.88	No
School B – School G	-0.8	0.15	0.01	Yes
School C – School D	0.45	0.15	0.05	Yes
School C – School E	−0.65	0.15	0.02	Yes
School C – School F	0.1	0.15	0.97	No
School C – School G	−0.55	0.15	0.03	Yes
School D – School E	−1.1	0.15	0.0005	Yes
School D – School F	−0.35	0.15	0.15	No
School D – School G	−1	0.15	0.001	Yes
School E – School F	0.75	0.15	0.01	Yes
School E – School G	0.1	0.15	0.95	No
School F – School G	−0.65	0.15	0.02	Yes

Teachers reported multiple constraints in integrating sustainability into numeracy lessons. Twelve teachers mentioned curriculum rigidity, ten cited limited instructional resources, and eight highlighted time constraints.

1. Curriculum Rigidity.

- "The curriculum is already packed with competencies we must achieve. Adding sustainability content feels like an extra burden rather than an integrated part." (Teacher 7).
- "We have little flexibility in the weekly schedule, so I focus on core math topics." (Teacher 2).

2. Limited Instructional Resources.

- "I never find word problems in the textbook that connect with green economy. I have to make my own, but it takes time and creativity." (Teacher 5).
- "I would use multimedia or local examples if they were provided, but most materials are generic." (Teacher 10).

3. Time Constraints.

- "We want to try new approaches, but preparing students for national exams consumes most of the time." (Teacher 9).
- "Inquiry-based lessons take longer than drills, so I rarely have time to implement them fully." (Teacher 14).

Theme 3: Expectations for Professional Development and Support

All 15 teachers indicated interest in structured professional development. Suggestions included workshops, peer learning, and ready-to-use teaching modules. Eleven teachers emphasized the need for institutional support.

Illustrative quotes:

- "If there is structured training with ready-to-use lesson plans, we will be more confident. It is not that we reject new ideas, but we need guidance to apply them properly." (Teacher 11).
- "Our effort will not last if there is no support from the principal or the district education office. It has to be a school-wide movement." (Teacher 1).
- "Collaborating with colleagues would help me adapt my lessons to include sustainability." (Teacher 6).

The interviews reveal the distribution of responses across the three themes: numeracy perceptions were mentioned by 13 teachers (with five linking it to sustainability), barriers were reported by 12–8 teachers depending on the type, and professional development needs were reported by all 15 teachers. The data provide descriptive insight into teachers' current practices, constraints, and support needs, without interpretation or discussion.

4.4 Classroom observation findings

A total of eight classroom observations were conducted across different primary schools to document how teachers integrate numeracy, sustainability, and green economy principles into mathematics instruction. Observations used a structured checklist covering

Table 3 Scoring definitions for implementation levels

Category	Score range	Criteria	Example
High implementation	8–10	Explicit integration of numeracy & green economy; student-centered; consistently applied	Project on household electricity consumption connecting costs to sustainability
Moderate implementation	5–7	Partial/inconsistent integration; mix of strategies	Sustainability occasionally mentioned, numeracy connections sporadic
Low implementation	0–4	Minimal or no integration; mainly teacher-centered	Rote arithmetic drills; sustainability only as side comment

Table 4 Observation results by indicator ($N = 8$)

Indicator	High Implementation (*n*)	Moderate (*n*)	Low (*n*)	Examples/Notes
Lesson design with sustainability context	2	3	3	Only 2 teachers embedded sustainability in lesson objectives (e.g., waste sorting as a basis for fractions).
Teaching strategies (student-centered)	1	4	3	Majority relied on lectures; only 1 teacher used inquiry-based activities.
Media and resources	2	3	3	Few used contextual media (e.g., recyclable materials for geometry shapes).
Student engagement	3	3	2	Engagement was higher when tasks related to students' everyday life (e.g., household electricity use).
Integration of numeracy with green economy	1	2	5	Integration was mostly implicit; only 1 teacher explicitly connected savings, waste, and cost efficiency.

five indicators: lesson design, teaching strategies, media and resources, student engagement, and integration of numeracy with green economy. Each indicator was scored using three levels (see Table 3). The aggregated observation results are presented in Table 4.

- High Implementation: Observable and explicit integration of sustainability or green economy in lesson objectives, activities, and assessment.
- Moderate Implementation: Partial or occasional integration; sustainability or green economy mentioned but not consistently linked to lesson outcomes.
- Low Implementation: Minimal or implicit integration; sustainability or green economy appears only as side comments or examples.

Observations showed that sustainability and green economy principles were rarely incorporated systematically. In most lessons, references to these topics were brief and incidental. For instance, one teacher mentioned farming when discussing area calculation, but this was not connected to broader sustainability themes. Direct instruction and textbook exercises dominated classroom practice, with isolated examples of contextualized or project-based activities. Notable examples included:

- A project-based lesson measuring household waste to calculate fractions.
- Using recycled bottles to measure volume in geometry exercises.

Among the eight classrooms, only one classroom demonstrated explicit integration of numeracy with green economy principles, while other classrooms showed partial or minimal integration. Student engagement tended to be higher in lessons that incorporated real-life sustainability examples, such as energy conservation or waste management.

Table 5 Expert validation results of the GENIUS model

Aspect	Indicator	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5	Expert 6	Expert 7	Expert 8	Aiken's V	Category
Content	Alignment with ESD	4	4	3	4	4	3	4	4	0.92	Very Valid
Content	Numeracy & green economy integration	4	4	4	3	4	4	4	4	0.88	Very Valid
Construct	Coherence of instruction	3	4	3	4	3	4	3	4	0.85	Valid
Construct	Clarity of goals	4	4	4	4	4	4	4	4	0.90	Very Valid
Practicality	Classroom feasibility	3	4	4	4	3	4	3	4	0.84	Valid
Practicality	Supporting tools availability	3	3	4	4	3	4	3	4	0.82	Valid

Threshold for high validity = Aiken's V ≥ 0.80 [1]

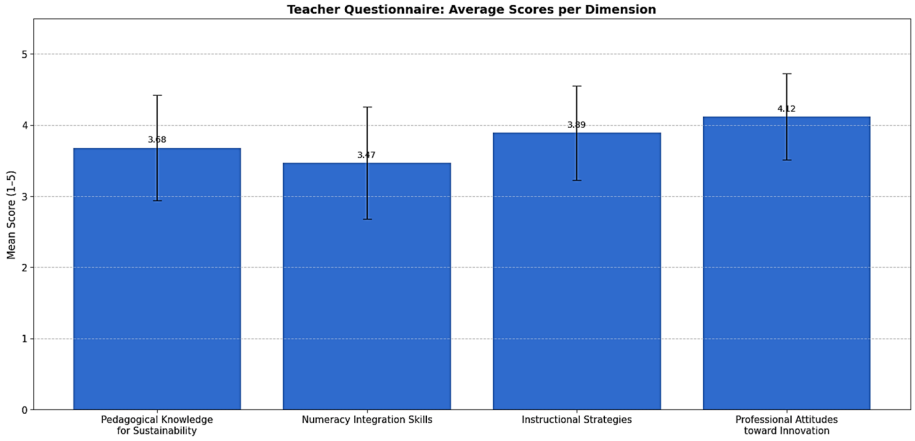


Fig. 2 Teacher questionnaire: Average scores per dimension

Overall, the classroom observations provide descriptive evidence of variability in practice across schools. The scoring criteria for High, Moderate, and Low implementation clarify the degree of sustainability integration in each observed lesson. These results can serve as a baseline for evaluating the impact of the GENIUS Model and for planning targeted interventions in professional development.

4.5 Expert validation results

Following the design of the GENIUS model, expert validation was conducted to assess the theoretical soundness, content validity, practicality, and classroom applicability of the model. A panel of eight experts participated: three mathematics education specialists, two ESD and green economy experts, and three senior primary school teachers with extensive teaching and curriculum design experience. Validation employed both quantitative rating sheets (using a four-point Likert scale, later analyzed with Aiken's V) and qualitative written comments gathered and individual reviews (see Table 5).

Figure 2 illustrates the average teacher scores across four questionnaire dimensions. Professional attitudes toward innovation scored highest (M = 4.12), indicating strong motivation and openness to adopting innovative pedagogical practices. In contrast, numeracy integration skills were lowest (M = 3.47), revealing challenges in practically embedding numeracy within sustainability contexts. Instructional strategies (M = 3.89) and pedagogical knowledge for sustainability (M = 3.68) fell at moderate levels,

suggesting adequate theoretical understanding and strategy use, yet highlighting the need for targeted support. Overall, the figure underscores a gap between teachers' motivation and their practical capacity, reinforcing the relevance of structured interventions such as the GENIUS Model.

In addition, the network diagram of interview themes and subthemes (see Fig. 3) visually maps the conceptual structure of teachers' perspectives regarding numeracy integration within Education for Sustainable Development (ESD). The diagram illustrates three central themes namely Perceptions of Numeracy within ESD, Barriers to Pedagogical Implementation, and Expectations for Professional Development, each connected to their respective subthemes. The visualization highlights how teachers recognize numeracy as foundational yet acknowledge a limited connection to sustainability, face structural and resource constraints in implementing pedagogical innovations, and express a strong demand for structured training, peer learning, and institutional support. By organizing qualitative findings into a hierarchical network, the figure clarifies the relationships among key themes, emphasizes areas of convergence and potential intervention, and enhances the accessibility of complex qualitative data for readers.

Quantitative Results

All indicators achieved Aiken's V scores above the 0.80 threshold, indicating that experts generally considered the GENIUS model to be valid. The content dimension scored highest (0.88–0.92), confirming alignment with ESD principles and the integration of numeracy and green economy. Construct validity ratings ranged from 0.85 to 0.90, showing that experts judged the instructional structure and clarity of learning goals as acceptable. Practicality indicators ranged from 0.82 to 0.84, suggesting feasibility of classroom implementation.

Distribution of Expert Ratings.

Although Aiken's V indicates high validity, individual ratings varied slightly. For instance:

- Integration of numeracy and green economy: 5 experts rated "4" (very relevant), 3 rated "3" (relevant).
- Feasibility of classroom implementation: 4 experts rated "4," 3 rated "3," and 1 rated "2" (moderately feasible).

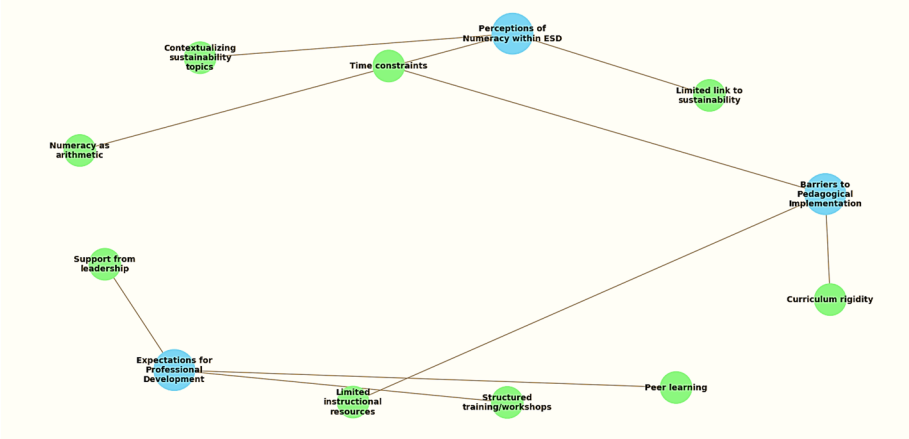


Fig. 3 Network diagram of interview themes and subthemes

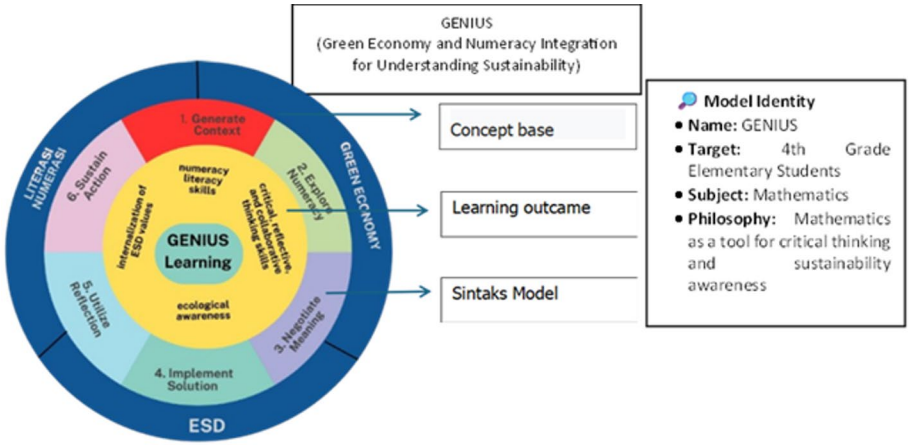


Fig. 4 GENIUS learning model

This shows overall agreement but also highlights minor variation in expert judgment, providing transparency regarding the robustness of the validation.

Qualitative Results.

Expert comments provided additional context:

- Content Experts: Emphasized the importance of linking sustainability topics to local contexts, such as waste management in Bogor.
- Mathematics Specialists: Recommended ensuring mathematical rigor is maintained while integrating ESD topics.
- Practicing Teachers: Suggested providing concrete lesson plans, worksheets, and teaching media to support classroom implementation. Several teachers recommended simple digital tools (e.g., interactive slides, calculators) to facilitate learning.

Overall, expert validation demonstrated that the GENIUS model is theoretically sound, relevant, and generally feasible for classroom use. Both quantitative and qualitative data indicate high agreement among experts on content, construct, and practicality, with minor variation in specific ratings. These results provide a descriptive foundation for further application of the model in classroom settings.

The subsequent section presents the GENIUS learning model framework, which integrates numeracy literacy, ecological awareness, and green economy principles for primary mathematics instruction (Fig. 4).

The GENIUS learning model is specifically designed for 4th-grade students in mathematics. This model integrates numeracy literacy, ecological awareness, and green economy principles, enabling students not only to learn mathematical concepts but also to understand their application in sustainable daily life. At its core, the model emphasizes the development of numeracy literacy, critical and reflective thinking skills, and ecological awareness as a foundation for learning that is relevant to global challenges.

The learning process in the GENIUS model is divided into six interconnected stages. The first stage, Generate Context, focuses on identifying relevant problems or contexts. The second stage, Explore Numeracy, involves exploring numerical concepts to understand sustainability issues, followed by Negotiate Meaning, which builds shared understanding through discussion. The next stages, Implement Solution, emphasizes

applying the designed solutions, then Utilize Reflection, reflecting on learning experiences and outcomes, and finally Sustain Action, which continues the sustainable actions undertaken.

The outer layer of the model connects learning with broader concepts such as literacy and numeracy, the green economy, and Education for Sustainable Development (ESD). The philosophy of the GENIUS model is that mathematics is not merely numbers or formulas but a tool for critical and reflective thinking and for fostering ecological awareness. Therefore, students are equipped not only with academic skills but also with the ability to understand environmental and social issues and act sustainably.

5 Discussion

5.1 Contextual needs and challenges in integrating sustainability into primary mathematics

The findings reveal that primary mathematics teachers in Bogor City are motivated to incorporate sustainability principles but face significant conceptual, structural, and contextual constraints. Needs analysis, questionnaires, interviews, and classroom observations consistently indicate that sustainability and green economy principles are rarely embedded systematically in lessons. Teachers often mention sustainability only as side notes, rather than integrating it into core numeracy tasks. Structural constraints—such as rigid curriculum frameworks, limited instructional time, and insufficient contextually relevant teaching resources—further impede implementation. Psychological and cultural factors may also contribute; for example, teachers prioritize exam-oriented tasks, reflecting systemic assessment pressures and traditional beliefs about mathematics as a purely numerical subject. These findings echo Pérez-Rodríguez et al. (2023), who emphasize insufficient professional development, and Hellberg & Knutsson (2022), who note gaps in teachers' conceptual understanding of ESD. Collectively, the data underscore the need for interventions that provide structured scaffolding, practical resources, and guidance to help teachers translate motivation into classroom practice. Importantly, classroom observation revealed that one school successfully integrated numeracy with green economy principles through a project-based lesson on household energy use, using recycled materials to calculate costs and environmental impact. This example illustrates that effective integration is feasible when teachers are supported with concrete tools and experiential activities.

5.2 Designing a mathematics learning model to embed green economy concepts

The study demonstrates that a structured mathematics learning model can systematically integrate green economy principles within the ESD framework. The GENIUS model, developed in response to the identified needs, organizes numeracy learning around real-life contexts, reflective practices, and project-based tasks connecting mathematics with sustainability. This design aligns with the broader literature on learning models, which function as conceptual frameworks structuring learning experiences, resources, and social interactions [23, 48]. GENIUS incorporates flexible scaffolding, multimodal learning tools, and context-specific examples to accommodate diverse learners, echoing Nel and Bornman [36] and Kostaki & Linardakis [24]. By linking numeracy to local sustainability issues such as waste management and energy conservation, the model operationalizes ESD principles in a manner that is both practical and pedagogically sound,

consistent with recommendations by Fernando and Tajan [14] and Giangrande et al., [16]. Compared with existing approaches (e.g., Bulut & Borromeo Ferri's modeling strategies or Lafuente-Lechuga et al.,'s integration methods), GENIUS uniquely combines project-based learning, reflection, and local contextualization to bridge motivation–practice gaps.

5.3 Validity of the GENIUS model for classroom implementation

Expert validation confirms that the GENIUS model is theoretically robust and practically feasible. Quantitative Aiken's V scores were high (0.82–0.92) across content, construct, and practicality dimensions, indicating strong agreement on the model's relevance and coherence. *Individual expert ratings varied slightly*, for instance in "numeracy and green economy integration," where five experts rated "4" and three rated "3," demonstrating minor differences in judgment but no extreme outliers. Practicality ratings also showed variation (one score of 2), highlighting areas needing additional guidance. Qualitative feedback emphasized linking content to local contexts, maintaining mathematical rigor, and providing ready-to-use lesson plans, worksheets, and media. These insights reflect the intention–practice gap: although teachers are motivated, lessons observed were largely teacher-centered and decontextualized. GENIUS addresses this by providing structured scaffolds, student-centered activities, and contextualized projects, enabling teachers to translate conceptual motivation into practice. This aligns with the multidimensional view of learning models that integrate pedagogy, social systems, and resources to foster meaningful and adaptive learning [19, 49].

5.4 Implications for theory, practice, and future research

The validated GENIUS model demonstrates that learning models can operationalize ESD and green economy principles effectively in primary mathematics classrooms. It bridges motivational readiness and practical competence, providing teachers with clear guidance and instructional tools. The model aligns with contemporary understandings of learning models as frameworks organizing strategies, interactions, and resources while remaining flexible for diverse learners [23, 48]. Embedding contextualized sustainability issues operationalizes UNESCO's vision of ESD as transformative education that equips learners with the knowledge, skills, and attitudes to act sustainably [17, 18].

Additionally, the discussion highlights critical limitations and avenues for future research. Methodological limitations, including reliance on self-report questionnaires, non-random sampling, and absence of longitudinal classroom testing, suggest the need for further empirical validation. Future studies could examine GENIUS implementation across different regions, grade levels, and socio-economic contexts, as well as compare its efficacy to other integration models. Exploring teacher beliefs, cultural influences, and assessment pressures could elucidate underlying factors contributing to the intention–practice gap. Finally, investigating the impact of progressive scaffolding and peer-collaborative structures on student outcomes would strengthen the evidence base for scaling ESD-focused mathematics interventions.

6 Conclusion

This study investigated the integration of sustainability principles into primary mathematics education in Indonesia and examined the design and validation of the GENIUS learning model. The findings indicate that while teachers demonstrate positive attitudes toward innovation and sustainability, actual classroom practices reveal limited and inconsistent integration of ESD and green economy concepts. Structural constraints including rigid curricula, limited resources, and insufficient professional development contribute to this intention–practice gap, highlighting the need for systematic pedagogical interventions.

In response, the GENIUS model was developed to embed green economy principles within a structured ESD framework, connecting numeracy learning to real-life contexts, reflective practices, and student-centered projects. Expert validation shows high content and construct validity, while practicality is supported through recommendations for ready-to-use lesson plans, scaffolding, and contextually relevant teaching media. It is important to note that the GENIUS model has not yet been empirically tested in large-scale classroom settings; current validation reflects expert consensus rather than direct student outcomes.

The study demonstrates that structured learning models like GENIUS can bridge the gap between teacher motivation and classroom implementation, providing both theoretical grounding and practical guidance for sustainability-oriented pedagogy. By integrating local sustainability issues, flexible scaffolding, and multimodal resources, the model offers a pathway for transforming traditional mathematics instruction into a platform for fostering environmental awareness, numeracy literacy, and sustainable citizenship. Importantly, these findings contribute to broader discussions in learning model research and ESD scholarship, suggesting that contextually adapted frameworks can support both teacher development and educational policy innovation. Future research should prioritize empirical testing through longitudinal and large-scale classroom trials, examining student outcomes and adapting the model to diverse subjects and educational contexts.

6.1 Recommendations and implications

Based on this study's findings, the following recommendations are proposed, with prioritization for practical implementation:

1. Teacher Professional Development (Highest Priority):

- Provide structured training on embedding ESD and green economy concepts in mathematics, including step-by-step use of the GENIUS model.
- Implement peer mentoring and collaborative planning to ensure consistency and innovation in practice.
- Emphasize student-centered strategies, such as project-based and inquiry-based learning, to foster engagement, critical thinking, and sustainability literacy.

2. School and Policy Support:

- Ensure provision of teaching media, low-cost technology, and access to local sustainability data.
- Introduce curricular flexibility to allow context-based projects reflecting local environmental, social, and economic issues.

- Incentivize schools that demonstrate successful integration of sustainability principles to encourage pedagogical innovation.

3. Curriculum Design:

- Explicitly embed sustainability and numeracy integration into learning standards and assessment guidelines.
- Utilize frameworks like GENIUS to align mathematical rigor with ESD competencies.
- Incorporate multi-level scaffolding and differentiated pathways to address diverse learner profiles and developmental stages.

4. Future Research Directions:

- Conduct longitudinal classroom trials to assess the impact of GENIUS on teacher practices and student learning outcomes.
- Compare implementation across regions, school types, and subjects to evaluate scalability and adaptability.
- Explore interdisciplinary applications, extending sustainability integration beyond mathematics.

Overall, these recommendations highlight the critical importance of connecting theoretical knowledge with practical application in ESD. Structured learning models such as GENIUS can enable teachers to overcome structural and skill-based barriers, fostering learners who possess both numeracy literacy and sustainability awareness. By explicitly acknowledging the *current developmental status of the GENIUS model* and focusing on prioritized interventions, this study advances both educational practice and theory, contributing to the broader goals of sustainable development, including environmental stewardship, social equity, and economic responsibility.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s43621-026-03056-5.

Supplementary Material 1

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Author contributions
Ratih Purnamasari led the conceptualization and overall coordination of the study; Nurlinda Safitri contributed to the research design and literature review; Siti Inayatul Faizah were responsible for data collection and analysis; Rumini Handayani assisted in data interpretation and visualization; Muhamad Alif Pratama contributed to drafting and editing the manuscript; All authors read and approved the final version of the manuscript.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the Research Ethics Committee of Universitas Pakuan, Bogor with approval number 142/UMJ/III/2025. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki. The research was conducted in accordance with the ethical guidelines and regulations established by the aforementioned ethics committee. Informed consent was obtained from all participants involved in this study. For participants under the age of 18, consent was obtained from a parent and/or legal guardian.

Consent to publication
Not applicable.

Clinical trial number
Not applicable.

Competing interests
The authors declare no competing interests.

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